
Global CO₂ Emissions Who Emits, How Much, and What's Changing

Fossil fuels & industry, 1750–2024

DATA SOURCE

Global Carbon Budget 2025
via Our World in Data

AUDIENCE

**Policy analysts, climate negotiators,
sustainability researchers**



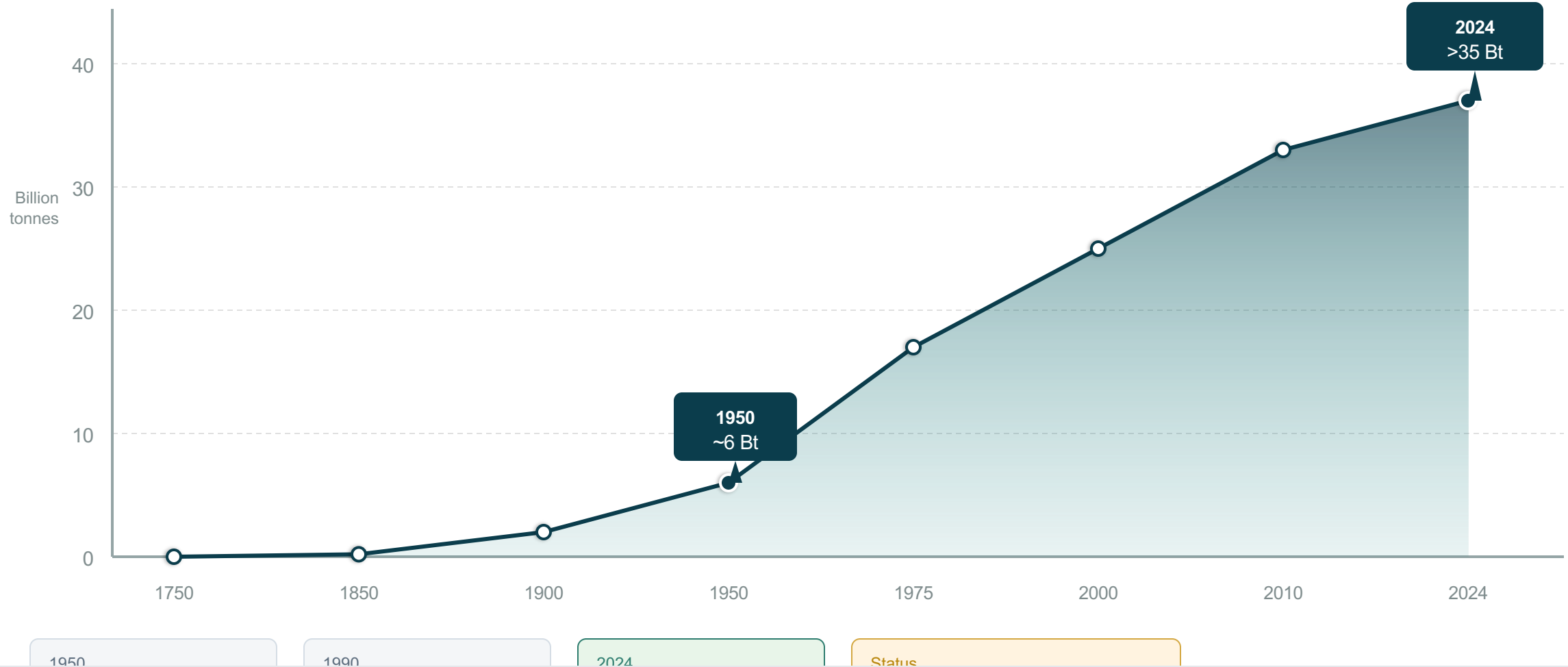
CONFIDENTIAL

02 Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Global Trajectory

KEY INSIGHT

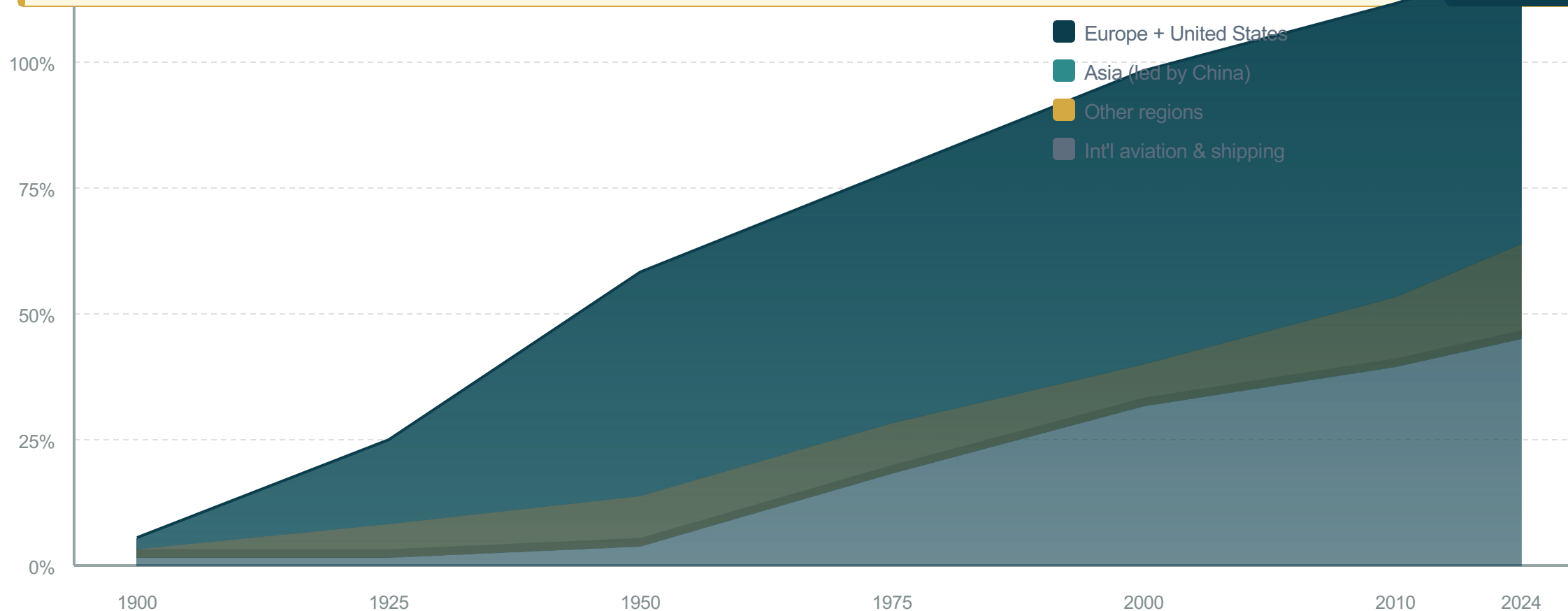
Growth has slowed in recent years, but emissions have not yet peaked



Source: Global Carbon Budget 2025 via Our World in Data

KEY INSIGHT

Asia — led by China — now dominates, producing roughly half of global emissions

2024
E+US <33%

KEY INSIGHT

China ~31%, US ~13%, India ~8%, and EU-27 ~6% together account for nearly 60% of global emissions

China
~31%

Rising, growth slowing

United States
~13%
Declining

Russia ~5%
Stable

Int'l transport
~3%

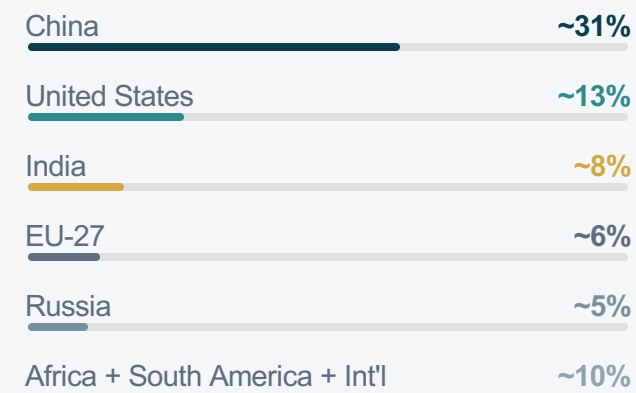
Africa
~3-4%
Slowly rising

S. America
~3-4%
Stable

Rest of
World
~27%

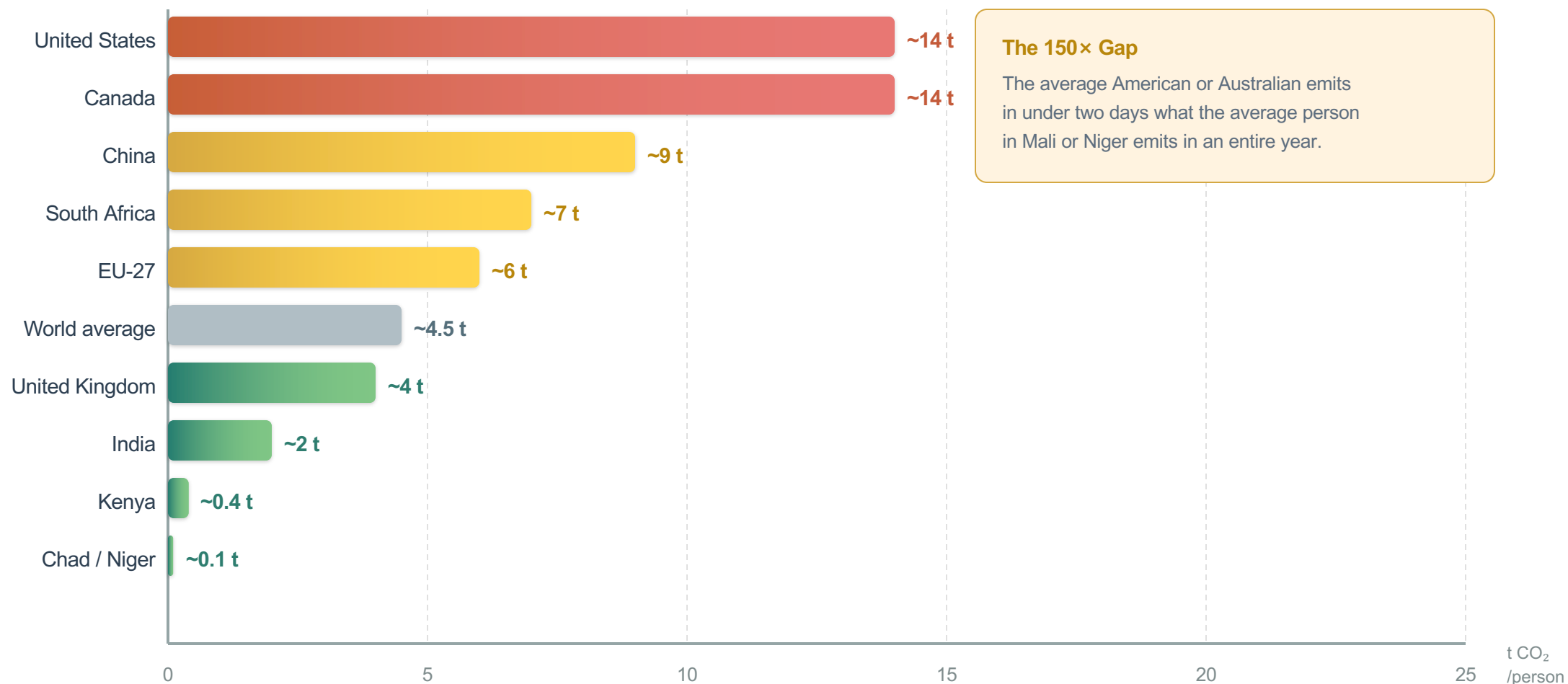
India
~8%
Rising

EU-27
~6%
Declining

Share of Global CO₂ (2024)

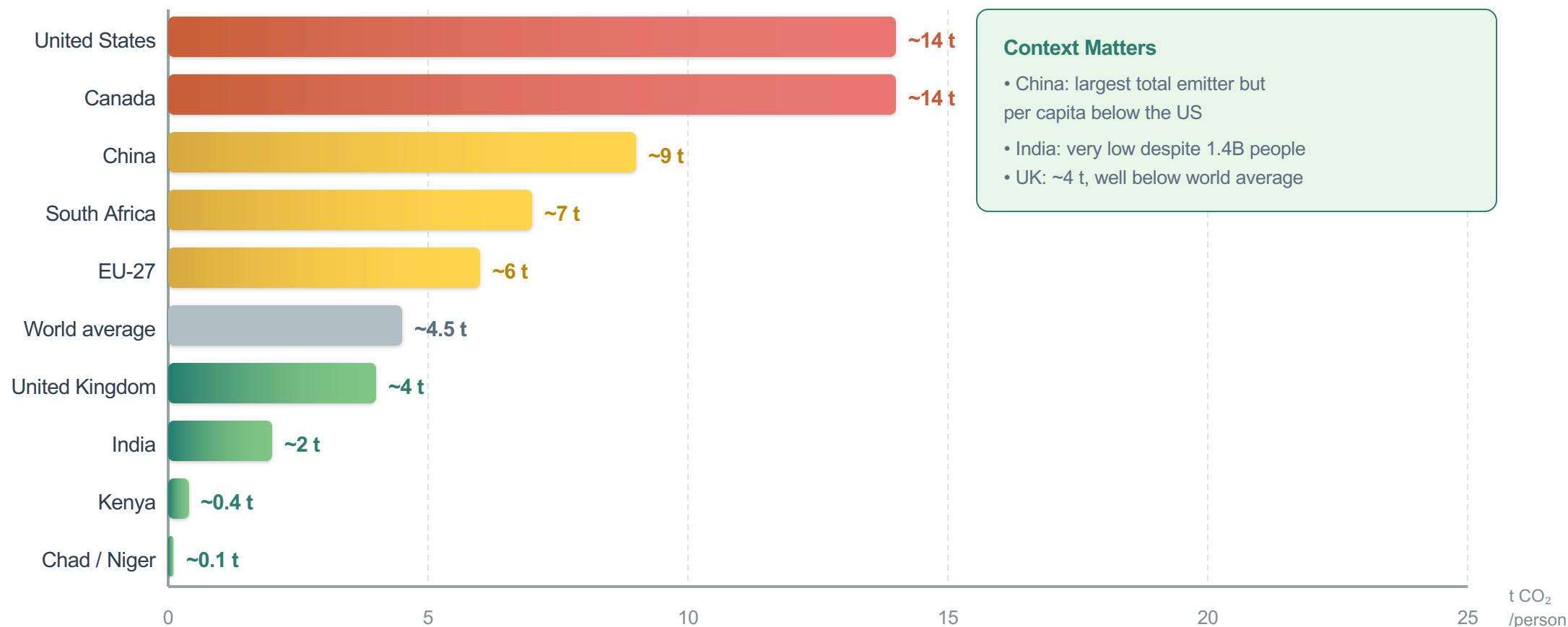
KEY INSIGHT

A ~150× gap exists between the lowest emitters (Sub-Saharan Africa) and highest large-country emitters



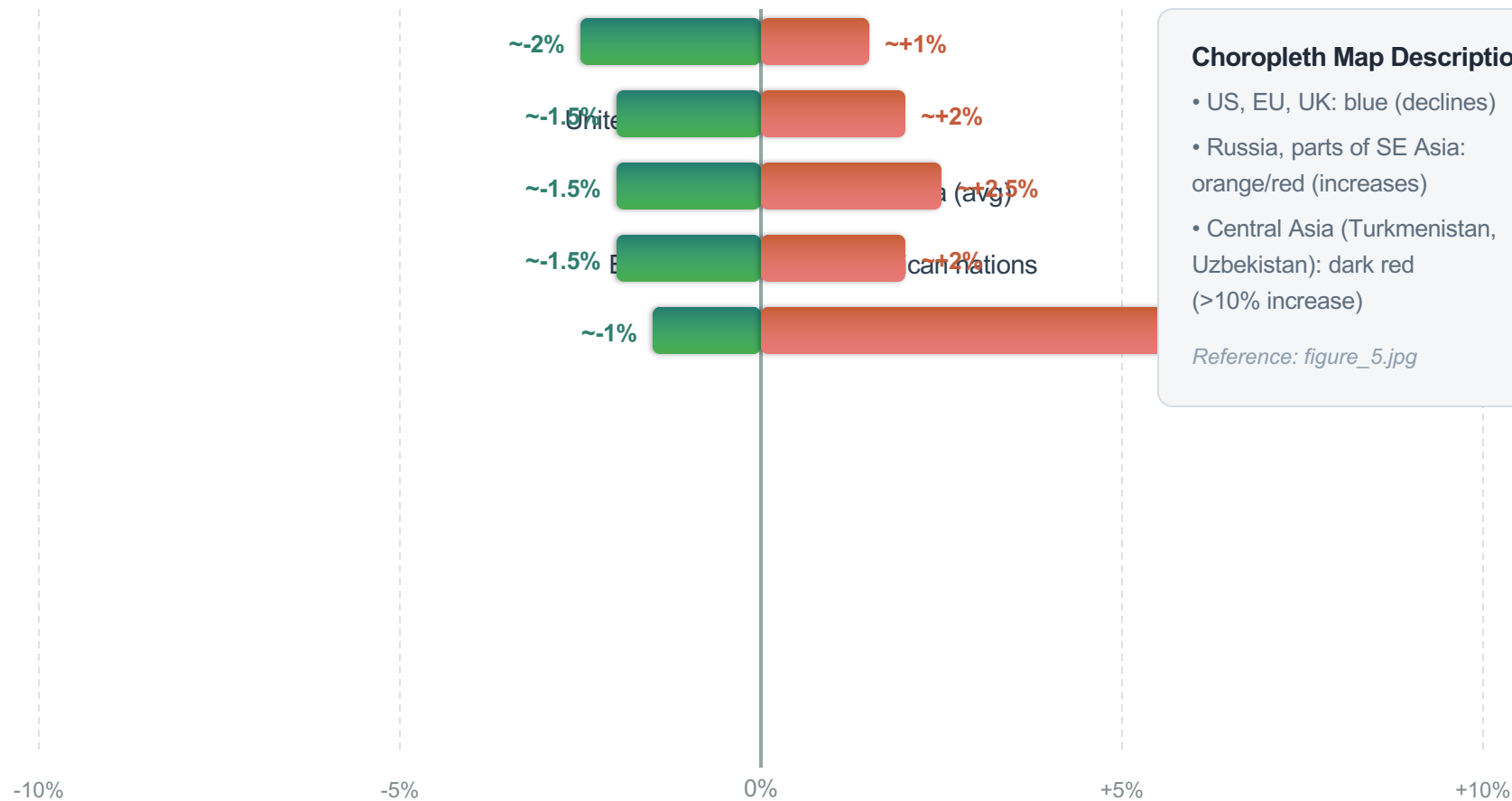
KEY INSIGHT

China's per capita remains below the US despite being the largest total emitter; India stays very low at ~2 t



KEY INSIGHT

Declines in the West were partially offset by rises in Russia, Southeast Asia, and parts of Africa



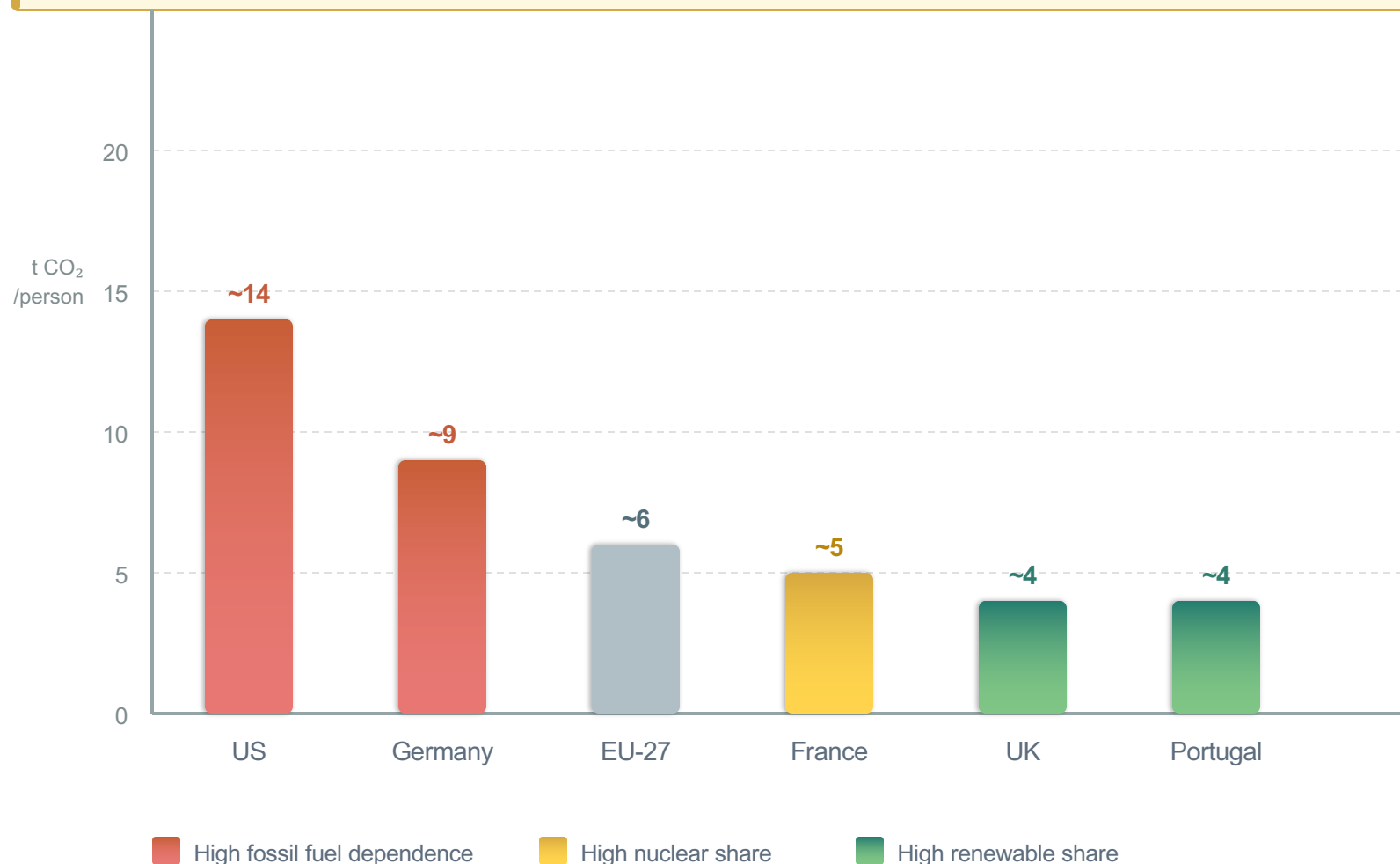
Choropleth Map Description

- US, EU, UK: blue (declines)
- Russia, parts of SE Asia: orange/red (increases)
- Central Asia (Turkmenistan, Uzbekistan): dark red (>10% increase)

Reference: figure_5.jpg

KEY INSIGHT

Countries with similar prosperity can have very different emissions depending on nuclear & renewable share



Why the Gap?

High Emitters

US, Germany, Netherlands:

- Heavy reliance on coal & gas
- Lower nuclear & renewable share
- Transportation fossil-dependent

Low Emitters

France, UK, Portugal:

- High nuclear (France: ~70%)
- Growing renewables (UK, Portugal)
- Decoupled prosperity from emissions

Policy Implication

Energy mix choices — not just wealth — drive per capita emissions.

1. NO PEAK YET

Global CO₂ emissions from fossil fuels exceed **35 billion tonnes per year** and have **not yet peaked**, despite slowing growth.

2. CHINA'S DUAL REALITY

China is the **largest annual emitter (~31%)** but per capita emissions (~9 t) remain **below the US (~14 t)**.

3. HISTORICAL RESPONSIBILITY

Europe and the US dominated emissions historically (**>90% in 1900**) but now account for **less than one-third**.

4. MASSIVE PER CAPITA INEQUALITY

Per capita gaps of **150×** persist between the richest (US/Canada ~14 t) and poorest nations (Chad/Niger ~0.1 t).

5. ENERGY POLICY MATTERS

Energy policy and technology choices can **decouple prosperity from emissions**— France and the UK prove it.